

APPENDIX

Competencies and Performance Indicators (PIs)

Computer Science & Engineering/Information Technology Programs

Appendix-A

PO 1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialisation for the solution of complex engineering problems.

Competency	Indicators
1.2 Demonstrate competence in mathematical modelling	1.2.1 Apply the knowledge of discrete structures, linear algebra, statistics and numerical techniques to solve problems 1.2.2 Apply the concepts of probability, statistics and queuing theory in modeling of computer-based system, data and network protocols.
1.5 Demonstrate competence in basic sciences	1.5.1 Apply laws of natural science to an engineering problem
1.6 Demonstrate competence in engineering fundamentals	1.6.1 Apply engineering fundamentals
1.7 Demonstrate competence in specialized engineering knowledge to the program	1.7.1 Apply theory and principles of computer science and engineering to solve an engineering problem

PO 2: Problem analysis: Identify, formulate, research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

Competency	Indicators
2.1 Demonstrate an ability to identify and formulate complex engineering problem	2.5.1 Evaluate problem statements and identifies objectives 2.5.2 Identify processes/modules/algorithms of a computer-based system and parameters to solve a problem 2.5.3 Identify mathematical algorithmic knowledge that applies to a given problem
2.6 Demonstrate an ability to formulate a solution plan and methodology for an engineering problem	2.6.1 Reframe the computer-based system into interconnected subsystems 2.6.2 Identify functionalities and computing resources. 2.6.3 Identify existing solution/methods to solve the problem, including forming justified approximations and assumptions 2.6.4 Compare and contrast alternative solution/methods to select the best methods 2.6.5 Compare and contrast alternative solution processes to select the best process.
2.7 Demonstrate an ability to formulate and interpret a model	2.7.1 Able to apply computer engineering principles to formulate modules of a system with required applicability and performance. 2.7.2 Identify design constraints for required performance criteria.
2.8 Demonstrate an ability to execute a solution process and analyze results	2.8.1 Applies engineering mathematics to implement the solution. 2.8.2 Analyze and interpret the results using contemporary tools. 2.8.3 Identify the limitations of the solution and sources/causes. 2.8.4 Arrive at conclusions with respect to the objectives.

PO 3: Design/Development of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety, and cultural, societal, and environmental considerations.

Competency	Indicators
3.5 Demonstrate an ability to define a complex/open-ended problem in engineering terms	3.5.1 Able to define a precise problem statement with objectives and scope. 3.5.2 Able to identify and document system requirements from stake- holders. 3.5.3 Able to review state-of-the-art literature to synthesize system requirements. 3.5.4 Able to choose appropriate quality attributes as defined by ISO/IEC/IEEE standard. 3.5.5 Explore and synthesize system requirements from larger social and professional concerns. 3.5.6 Able to develop software requirement specifications (SRS).
3.6 Demonstrate an ability to generate a diverse set of alternative design solutions	3.6.1 Able to explore design alternatives. 3.6.2 Able to produce a variety of potential design solutions suited to meet functional requirements. 3.6.3 Identify suitable non-functional requirements for evaluation of alternate design solutions.
3.7 Demonstrate an ability to select optimal design scheme for further development	3.7.1 Able to perform systematic evaluation of the degree to which several design concepts meet the criteria. 3.7.2 Consult with domain experts and stakeholders to select candidate engineering design solution for further development
3.8 Demonstrate an ability to advance an engineering design to defined end state	3.8.1 Able to refine architecture design into a detailed design within the existing constraints. 3.8.2 Able to implement and integrate the modules. 3.8.3 Able to verify the functionalities and validate the design.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

Competency	Indicators
4.4 Demonstrate an ability to conduct investigations of technical issues consistent with their level of knowledge and understanding	4.4.1 Define a problem for purposes of investigation, its scope and importance 4.4.2 Able to choose appropriate procedure/algorithm, dataset and test cases. 4.4.3 Able to choose appropriate hardware/software tools to conduct the experiment.
4.5 Demonstrate an ability to design experiments to solve open-ended problems	4.5.1 Design and develop appropriate procedures/methodologies based on the study objectives
4.6 Demonstrate an ability to analyze data and reach a valid conclusion	4.6.1 Use appropriate procedures, tools and techniques to collect and analyze data 4.6.2 Critically analyze data for trends and correlations, stating possible errors and limitations 4.6.3 Represent data (in tabular and/or graphical forms) so as to facilitate analysis and explanation of the data, and drawing of conclusions 4.6.4 Synthesize information and knowledge about the problem from the raw data to reach appropriate conclusions

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

Competency	Indicators
5.4 Demonstrate an ability to identify/create modern engineering tools, techniques and resources	5.4.1 Identify modern engineering tools, techniques and resources for engineering activities 5.4.2 Create/adapt/modify/extend tools and techniques to solve engineering problems
5.5 Demonstrate an ability to select and apply discipline-specific tools, techniques and resources	5.5.1 Identify the strengths and limitations of tools for (i) acquiring information, (ii) modeling and simulating, (iii) monitoring system performance, and (iv) creating engineering designs. 5.5.2 Demonstrate proficiency in using discipline-specific tools
5.6 Demonstrate an ability to evaluate the suitability and limitations of tools used to solve an engineering problem	5.6.1 Discuss limitations and validate tools, techniques and resources 5.6.2 Verify the credibility of results from tool use with reference to the accuracy and limitations, and the assumptions inherent in their use.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal, and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

Competency	Indicators
6.3 Demonstrate an ability to describe engineering roles in a broader context, e.g. pertaining to the environment, health, safety, legal and public welfare	6.3.1 Identify and describe various engineering roles; particularly as pertains to protection of the public and public interest at the global, regional and local level
6.4 Demonstrate an understanding of professional engineering regulations, legislation and standards	6.4.1 Interpret legislation, regulations, codes, and standards relevant to your discipline and explain its contribution to the protection of the public

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and the need for sustainable development.

Competency	Indicators
7.3 Demonstrate an understanding of the impact of engineering and industrial practices on social, environmental and in economic contexts	7.3.1 Identify risks/impacts in the life-cycle of an engineering product or activity 7.3.2 Understand the relationship between the technical, socio-economic and environmental dimensions of sustainability
7.4 Demonstrate an ability to apply principles of sustainable design and development	7.4.1 Describe management techniques for sustainable development 7.4.2 Apply principles of preventive engineering and sustainable development to an engineering activity or product relevant to the discipline

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

Competency	Indicators
8.3 Demonstrate an ability to recognize ethical dilemmas	8.3.1 Identify situations of unethical professional conduct and propose ethical alternatives

8.4	Demonstrate an ability to apply the Code of Ethics	8.4.1	Identify tenets of the ASME professional code of ethics
		8.4.2	Examine and apply moral & ethical principles to known case studies
PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.			
Competency		Indicators	
9.4	Demonstrate an ability to form a team and define a role for each member	9.4.1	Recognize a variety of working and learning preferences; appreciate the value of diversity on a team
		9.4.2	Implement the norms of practice (e.g. rules, roles, charters, agendas, etc.) of effective team work, to accomplish a goal.
9.5	Demonstrate effective individual and team operations--communication, problem-solving, conflict resolution and leadership skills	9.5.1	Demonstrate effective communication, problem-solving, conflict resolution and leadership skills
		9.5.2	Treat other team members respectfully
		9.5.3	Listen to other members
		9.5.4	Maintain composure in difficult situations
9.6	Demonstrate success in a team-based project	9.6.1	Present results as a team, with smooth integration of contributions from all individual efforts
PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with the society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions			
Competency		Indicators	
10.4	Demonstrate an ability to comprehend technical literature and document project work	10.4.1	Read, understand and interpret technical and non-technical information
		10.4.2	Produce clear, well-constructed, and well-supported written engineering documents
		10.4.3	Create flow in a document or presentation - a logical progression of ideas so that the main point is clear
10.5	Demonstrate competence in listening, speaking, and presentation	10.5.1	Listen to and comprehend information, instructions, and viewpoints of others
		10.5.2	Deliver effective oral presentations to technical and non-technical audiences
10.6	Demonstrate the ability to integrate different modes of communication	10.6.1	Create engineering-standard figures, reports and drawings to complement writing and presentations
		10.6.2	Use a variety of media effectively to convey a message in a document or a presentation
PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's work, as a member and leader in a team, to manage projects and in multidisciplinary environments.			
Competency		Indicators	
11.4	Demonstrate an ability to evaluate the economic and financial performance of an engineering activity	11.4.1	Describe various economic and financial costs/benefits of an engineering activity
		11.4.2	Analyze different forms of financial statements to evaluate the financial status of an engineering project
11.5	Demonstrate an ability to compare and contrast the costs/benefits of alternate proposals for an engineering activity	11.5.1	Analyze and select the most appropriate proposal based on economic and financial considerations.

11.6 Demonstrate an ability to plan/manage an engineering activity within time and budget constraints	11.6.1 Identify the tasks required to complete an engineering activity, and the resources required to complete the tasks. 11.6.2 Use project management tools to schedule an engineering project, so it is completed on time and on budget.
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PO 12: Life-long learning: Recognise the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Competency	Indicators
12.4 Demonstrate an ability to identify gaps in knowledge and a strategy to close these gaps	12.4.1 Describe the rationale for the requirement for continuing professional development 12.4.2 Identify deficiencies or gaps in knowledge and demonstrate an ability to source information to close this gap
12.5 Demonstrate an ability to identify changing trends in engineering knowledge and practice	12.5.1 Identify historic points of technological advance in engineering that required practitioners to seek education in order to stay current 12.5.2 Recognize the need and be able to clearly explain why it is vitally important to keep current regarding new developments in your field
12.6 Demonstrate an ability to identify and access sources for new information	12.6.1 Source and comprehend technical literature and other credible sources of information 12.6.2 Analyze sourced technical and popular information for feasibility, viability, sustainability, etc.

Batch: 2021-2025

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9) CS1201A - Fundamentals of Computing Laboratory	C1171	3.00	3.00	2.00	2	1	-	-	2.00	2.00	1	1.00	3.00	2.00
	C1172	3	3	3	2	2	-	-	2	2	2	2	2	2
	C1173	3	3	3	2	1	-	-	1	2	1	2	2	2
	C1174	3	3	2	2	2	-	-	1	2	2	2	2	2
	C1175	3	3	2	2	1	-	-	2	2	1	2	2	2
	C117	3	3	2.4	2	1	-	-	1.4	2	1	1.8	2.2	2
	HISEMSTER (2022-2023) ODD													
1) MA1303A - Discrete Mathematics	C2011	3	2	2	1	-	-	-	-	-	-	-	2	-
	C2012	3	2	2	1	-	-	-	-	-	-	-	1	-
	C2013	3	2	1	2	-	-	-	-	-	-	-	2	-
	C2014	3	2	1	1	-	-	-	-	-	-	-	2	-
	C2015	3	2	2	1	-	-	-	-	-	-	-	1	-
	C201	3.00	2.00	1.60	1.20	-	-	-	-	-	-	-	1.60	-
	C2021	2.00	3.00	2.00	3.00	-	-	-	3.00	-	-	2.00	2.00	2.00
2) IT1301A - Object Oriented Programming	C2022	2	2	2	3	2	-	-	2	-	-	2	2	2
	C2023	3	2	2	2	3	-	-	2	-	-	2	2	2
	C2024	2	2	2	2	3	-	-	3	-	-	2	2	2
	C2025	2	3	2	3	2	-	-	2	-	-	2	2	2
	C202	2.20	2.40	2.00	2.60	2.60	-	-	2.40	-	-	2.00	2.00	2.00
	C2031	3.00	2.00	2.00	-	-	-	-	-	-	-	2.00	2.00	2.00
	C2032	3	2	2	-	-	-	-	-	-	-	2	2	2
3) CS1401A - Data Structures	C2033	3	2	2	-	-	-	-	-	-	-	2	3	3
	C2034	3	2	2	-	-	-	-	-	-	-	2	2	2
	C2035	3	2	2	-	-	-	-	-	-	-	2	3	2
	C203	3.00	2.00	2.00	-	-	-	-	-	-	-	2.00	2.40	2.20
	C2041	3.00	3.00	3.00	3.00	3.00	-	-	1.00	-	-	2.00	2.00	1.00
	C2042	3	3	3	3	3	-	-	2	-	-	2	2	1
	C2043	3	3	3	3	3	-	-	2	-	-	2	2	1
4) IC1306A - Digital Principles and System Design	C2044	3	3	3	3	3	-	-	2	-	-	2	2	1
	C2045	3	3	3	3	3	-	-	3	-	-	2	2	1
	C204	3.00	3.00	3.00	3.00	3.00	-	-	2.20	-	-	2.00	2.00	1.00
	C2051	2.00	2.00	2.00	-	1.00	-	-	1.00	2.00	1.00	1.00	1.00	1.00
	C2052	2	2	2	-	1	-	-	1	2	1	1	1	1
	C2053	2	2	2	-	1	-	-	1	2	1	1	1	1
	C2054	2	2	2	-	1	-	-	1	2	1	1	1	1
5) CS1402 A- Software Engineering	C2055	2	2	2	-	1	-	-	1	2	1	1	1	1
	C205	2.00	2.00	2.00	-	1.00	-	-	1.00	2.00	1.00	1.00	1.00	1.00
	C205	2.00	2.00	2.00	-	1.00	-	-	1.00	2.00	1.00	1.00	1.00	1.00

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		C2171	3	2	1	3	-	2	3	1	3	3	3	2	-
		C2172	3	1	-	3	2	2	3	3	3	3	2	-	-
		C2173	3	1	-	3	-	1	3	1	3	3	2	-	-
		C2174	1	3	1	3	-	2	3	2	3	3	2	-	-
		C2175	2	1	2	3	3	1	3	2	3	3	3	-	-
		C217	2.4	2.0	1.3	3.0	2.5	1.6	3.0	1.8	3.0	3.0	1.8	-	-
181201 A-1 Implementation And 9) Self Study Laboratory															
A SEMESTER (2023-2024) ODD															
		C3011	3	3	3	2	-	-	-	-	-	-	-	2	-
		C3012	3	3	3	2	-	-	-	-	-	-	-	2	-
		C3013	3	3	3	2	-	-	-	-	-	-	-	2	-
		C3014	3	3	3	2	-	-	-	-	-	-	-	2	-
		C3015	3	3	3	2	-	-	-	-	-	-	-	2	-
		C301	3.00	3.00	2.00	-	-	-	-	-	-	-	-	2.00	-
		C3021	3.00	2.00	2.00	-	-	-	-	-	-	-	-	2.00	1.00
		C3022	2	2	2	-	-	-	-	-	-	-	-	2	1
		C3023	2	2	2	-	-	-	-	-	-	-	-	2	2
		C3024	2	2	2	1	-	-	-	-	-	-	-	2	1
		C3025	2	2	2	1	-	-	-	-	-	-	-	2	1
		C302	2.20	2.00	1.60	-	-	-	-	-	-	-	-	1.60	1.40
		C3031	3.00	2	1	-	-	-	-	-	-	-	-	2	2
		C3032	3	3	2	-	-	-	-	-	-	-	-	2	1
		C3033	3	3	2	-	-	-	-	-	-	-	-	2	2
		C3034	3	1	1	-	-	-	-	-	-	-	-	2	1
		C3035	3	2	1	-	-	-	-	-	-	-	-	2	2
		C303	3.00	2.20	1.40	-	-	-	-	-	-	-	-	2.00	1.40
		C3041	3.00	2.00	2.00	-	-	-	-	-	-	-	-	2.00	1.00
		C3042	2	1	1	-	-	-	-	-	-	-	-	2	1
		C3043	1	1	1	-	-	-	-	-	-	-	-	2	2
		C3044	1	2	2	-	-	-	-	-	-	-	-	2	1
		C3045	2	2	1	-	-	-	-	-	-	-	-	2	1
		C304	1.80	1.60	1.40	-	-	-	-	-	-	-	-	2.00	1.20
		C3051	3.00	3.00	3.00	-	-	-	-	-	-	-	-	3.00	1.00
		C3052	3	3	3	2	-	2	-	-	2	1	2	2	2
		C3053	3	3	3	-	-	1	-	-	2	1	2	2	2
		C3054	3	3	3	1	-	1	-	-	1	2	2	2	2
		C3055	3	3	3	2	-	2	-	-	2	2	2	2	2
		C305	3.00	3.00	3.00	1.67	1.50	-	-	-	1.75	1.50	2.00	1.80	1.80
4		CS1601A - Theory of Computation													
5		EC1515A - Embedded System													

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5	CS1607A - Cyber Security	C312.1	3.00	3.00	3.00	3.00	3.00	3.00	2.00	2.00	1.00	2.00	2.00	2.00	3.00	3.00
		C312.2	2	3	2	1	1	1	2	2	1	2	2	2	3	3
		C312.3	2	3	2	2	3				2	1	1	2	3	3
		C312.4	2	2	2	1	1	1	3	2	1	1	1	2	3	3
		C312.5	2	2	2	2	1	2	2	2	1	1	1	3	3	3
		C312	2.20	2.60	2.20	1.80	1.80	1.80	2.67	2.00	1.20	1.40	1.40	2.00	3.00	3.00
6	CS1608A - Software Testing	C313.1	3.00	2.00	3.00	2.00				1					2.00	2
		C313.2	3	3	2	3				2					2	1
		C313.3	2	2	3	2				2					2	1
		C313.4	3	3	2	2				1					2	2
		C313.5	2	2	3	2				1					2	1
		C313	2.60	2.40	2.60	2.20			1.40						2.00	1.40
7	CS1605A - Internet Programming Laboratory	C314.1	3.00	3.00	3.00	1.00	3.00			1.00	3.00	2.00	3.00	2.00	2.00	2.00
		C314.2	3	2	3	2	3			1	3	1	3	2	3	2
		C314.3	3	2	3	2	3			2	3	1	3	2	2	1
		C314.4	3	2	3	2	3			1	3	2	3	2	2	1
		C314.5	3	2	3	2	3			1	3	1	3	2	2	1
		C314	3.00	2.20	3.00	1.80	3.00			1.20	3.00	1.40	3.00	2.00	2.20	1.40
8	CS1606A - Security Laboratory	C315.1	3.00	2.00	2.00	2.00	2.00							2.00	3.00	2.00
		C315.2	3	3	2	2	1							2	3	3
		C315.3	3	2	2	2	2							2	3	3
		C315.4	3	3	3	2	2							2	2	3
		C315.5	3	3	3	2	1							2	2	3
		C315	3.00	2.60	2.40	2.00	1.60							2	2	3
9	IS1601A - Professional Communication Laboratory	C316.1					3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00		
		C316.2						3	3	3	3	3	3	3		
		C316.3					3	3	3	3	3	3	3	3		
		C316.4					3	3	3	3	3	3	3	3		
		C316.5					3	3	3	3	3	3	3	3		
		C316					3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00		

VISEM/INTER(2024-2025) ODD

1	IT1701A - Distributed Systems and Cloud Computing	C401.1	3	2	3				1	2	2	2	2	2	2	2
		C401.2	3	2	2				2	2	2	2	2	2	2	2
		C401.3	2	2	3				2	2	2	2	2	2	2	2
		C401.4	3	2	2				2	2	1	2	2	2	2	2
		C401.5	2	2	3				1	2	1	2	2	2	2	2
		C401	2.60	2.00	2.60				1.60	2.00	1.60	2.00	2.00	2.00	2.00	2.00

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	C410.1	2.00	2.00	3.00	2	3.00	-	2.00	-	3.00	2.00	3.00	2.00	2.00	2.00
	C410.2	3	2	3	2	3	-	2	-	3	2	3	2	2	2
	C410.3	2	2	3	2	3	-	2	-	3	2	3	2	2	2
	C410.4	3	2	3	2	3	-	2	-	3	2	3	2	2	2
	C410.5	2	2	3	2	3	-	2	-	3	2	3	2	2	2
IT1702A-Cloud Computing Laboratory	C410	2.40	2.00	3.00	2.00	3.00	-	2.00	-	3.00	2.00	3.00	2.00	2.00	2.00
	C408.1	2.0	2.0	2.0	1.0	1.0	-	-	-	-	1.0	-	1.0	1.0	2.0
	C408.2	2	2	2	1	1	-	-	-	-	1	-	1	1	2
	C408.3	2	2	2	1	1	-	-	-	-	1	-	1	1	2
	C408.4	2	2	2	1	1	-	-	-	-	1	-	1	1	2
	C408.5	2	2	2	1	1	-	-	-	-	1	-	1	1	2
CS1702A-Machine Learning Laboratory	C408	2.00	2.00	2.00	1.00	1.00	-	-	-	-	1.00	-	1.00	1.00	2.00
	C411.1	3.00	3.00	3.00	3.00	1.00	1.00	1.00	3.00	3.00	3.00	3.00	3.00	3.00	1.00
	C411.2	3	3	2	1	2	-	-	-	3	3	3	3	3	3
	C411.3	3	3	3	2	1	-	-	1	3	3	3	3	3	3
	C411.4	1	1	-	-	-	-	-	1	3	3	3	3	3	3
CS1714A-Project Work I	C411.5	3	3	3	3	2	1	1	1	3	2	-	2	3	2
	C411	2.60	2.60	2.75	2.25	1.50	1.00	1.00	1.50	3.00	2.80	3.00	3.00	2.75	2.25
	C409.1	2.00	1.00	2.00	2.00	3.00	-	-	-	1.00	1.00	-	-	3.00	1.00
	C409.2	2	1	2	2	3	3	3	1	1	1	1	2	3	1
	C409.3	2	3	3	3	3	-	-	-	3	3	2	2	3	3
	C409.4	2	3	3	3	3	3	3	2	3	3	2	3	2	2
	C409.5	2	3	3	3	3	3	3	3	3	3	2	3	2	1
CS1703A-Summer Internship	C409	2.00	2.20	2.60	2.60	3.00	3.00	3.00	2.60	2.30	2.20	1.75	2.50	2.60	1.60
VIII SEMESTER (2024-2025) EVEN															
	C412.1	-	-	-	-	-	2	-	3	3	-	-	2	1	-
	C412.2	-	-	-	-	-	-	2	3	2	1	-	1	1	-
	C412.3	-	-	-	-	-	-	-	3	-	-	-	-	1	-
	C412.4	-	-	-	-	-	2	2	3	2	1	-	3	1	-
	C412.5	-	-	-	-	-	2	2	3	1	2	-	2	1	-
GE1801A-Professional Ethics in Engineering	C412	-	-	-	-	-	2.00	2.00	3.00	2.00	1.33	-	2.25	1.00	-
	C413.1	2.00	2.00	2.00	2.00	-	-	-	-	-	-	-	-	2	1
	C413.2	2	1	1	2	-	-	-	-	-	-	-	-	2	1
	C413.3	2	3	3	2	-	-	-	-	-	-	-	-	2	2
	C413.4	3	1	2	3	-	-	-	-	-	-	-	-	1	2
	C413.5	3	1	1	3	-	-	-	-	-	-	-	-	1	2
CS1809A-UI & UX Design	C413	2.40	1.60	1.80	2.40	-	-	-	-	-	-	-	-	1.60	1.60

3	CS1814A-Ethical hacking	C414.1	2.00	2.00	2.00	2.00	1.00	2.00	3.00	-	2.00	1.00	2.00	2.00	2.00
		C414.2	2	2	1	1	2	2	2	-	-	1	1	1	2
		C414.3	3	2	2	2	1	2	1	-	1	1	1	2	2
		C414.4	2	2	1	1	-	2	-	-	-	-	2	2	2
		C414.5	3	2	2	2	2	-	-	-	-	-	1	2	2
		C418	2.4	2	1.6	2	1	2	2	-	2	1	1.4	2	2
		C415.1	2	1	2	-	3	-	-	-	3	-	1	1	1
4	CS1812A-Digital Marketing	C415.2	2	1	1	-	3	-	-	-	3	-	2	2	2
		C415.3	2	2	2	-	3	-	-	-	3	-	2	1	1
		C415.4	2	2	2	-	3	-	-	-	3	-	2	2	1
		C415.5	2	2	2	-	3	-	-	-	3	-	1	1	2
		C415	2	1.6	1.8	-	3	-	-	-	3	-	1.6	1.4	1.4
		C416.1	3	3	3	3	3	3	3	3	3	3	3	3	3
		C416.2	3	3	2	2	3	3	2	3	2	3	3	3	3
5	CS1801A- Project Work II	C416.3	3	3	3	2	2	3	3	2	3	3	2	3	3
		C416.4	3	3	3	2	3	3	3	2	2	1	3	3	3
		C416.5	3	3	2	1	3	2	2	1	2	2	2	3	3
		C416	3.0	3.0	2.6	2.0	2.8	2.6	2.6	2.2	2.4	2.4	2.4	3.0	3.0
		C416	3.0	3.0	2.5	1.8	2.8	2.5	2.5	2.0	2.3	2.3	2.3	3.0	3.0

Process Owner

HoD

Batch: 2020-2024

[illegible]

6	ME1101 - Engineering Graphics	C105.5	Implement file concepts to read and write data from/to files.																		1	2										
		C'105																		2.00	2.00											
		C106.1	2	3	3	2	2	3	1	3	3	3	1	2	2	1	1	1	2													
		C106.2	3	3	3	3	1	3	1	2	3	2	1	3	3	1	1	1	1													
		C106.3	2	3	3	2	1	3	1	2	1	2	1	2	1	1	1	1	1													
		C106.4	3	3	3	3	1	3	1	1	2	3	1	2	2	1	1	1	1													
7	DS1101 - Physics and Chemistry Laboratory	C106.5	3	3	3	3	1	3	1	2	3	3	1	2	3	1	1	1	1													
		C'106																		2.60	3.00	3.00	2.60	1.20	3.00	1.00	2.00	2.40	1.00	1.40	-	
		C107.1	2	1	1	1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								
		C107.2	2	1	1	1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								
		C107.3	2	1	1	1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								
		C107.4	2	2	1	1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								
8	CS1102 - Problem Solving and Python Programming Laboratory	C107.5	2	1	1	1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-								
		C'107																		2.00	1.20	1.00	1.00	-	-	-	-	-	-	-	-	-
		C108.1	3	3	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	1	-								
		C108.2	3	3	2	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	1	-								
		C108.3	3	3	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	1	-								
		C108.4	2	2	2	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	1	-								
8		C108.5	3	3	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-	-	2	1	-									
		C'108																		2.80	2.80	2.60	2.80	-	-	-	-	-	-	2.00	1.00	-

6	ME1101 - Engineering Graphics	C105.5	Implement file concepts to read and write data from/to files.																		3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
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	C115.1	Develop C programs for simple applications making use of basic constructs	3	2	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	2	2
	C115.2	Develop C programs for control statements	3	2	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	2	2
	C115.3	Develop C programs involving arrays, strings and pointers	3	2	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	2	2
	C115.4	Develop C programs involving functions, and structures	3	2	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	2	2
	C115.5	Design applications using sequential and random access file processing	3	2	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	2	2
		C115	3.00	2.00	3.00	2.00	2.00	-	-	-	-	-	-	-	-	-	-	-	-	2.00	2.00	2.00
	C116.1	Understand the basic principle of welding	3	3	3	2	2	3	-	3	2	3	-	1	-	-	-	-	-	-	-	-
	C116.2	Understand the basic principle of Sheet Metal Work	3	3	3	2	2	3	-	3	2	3	-	1	-	-	-	-	-	-	-	-
	C116.3	Understand the basic principle of joints	3	3	3	2	2	3	-	3	2	3	-	1	-	-	-	-	-	-	-	-
	C116.4	Understand the basic principle of Machine assembly practice	3	3	3	2	2	3	-	3	2	3	-	1	-	-	-	-	-	-	-	-
	C116.5	Understand the basic principle of Drilling	3	3	3	2	2	3	-	3	2	3	-	1	-	-	-	-	-	-	-	-
		C116	3.00	3.00	3.00	2.00	2.00	3.00	-	3.00	2.00	3.00	-	1.00	-	-	-	-	-	-	-	-
	C117.1	Use MS-Office for designing a Document, Presentations and Excel sheet.	3	3	2	2	1	-	-	1	2	2	1	1	3	2	-	-	-	2	2	2
	C117.2	Design and develop static web pages by using the Mark up languages that meet the specified needs and interests.	3	3	3	2	2	-	-	2	2	2	1	2	2	2	-	-	-	2	2	2
	C117.3	Understand the Networking Concepts.	3	3	3	2	1	-	-	1	2	2	1	2	2	2	-	-	-	2	2	2
	C117.4	Use typical data definitions and manipulation commands.	3	3	2	2	2	-	-	1	2	2	1	2	2	2	-	-	-	2	2	2
	C117.5	Assemble Computer Hardware and Install software.	3	3	2	2	1	-	-	2	2	2	1	2	2	2	-	-	-	2	2	2
		C117	3	3	2.4	2	1.4	-	-	1.4	2	2	1	1.8	2.2	-	-	-	-	2.2	2	2

THESEMASTER(2021-2022)0000

[illegible]

5	CS1302 - Software Engineering	C205.1	Develop the project by adopting suitable lifecycle models.	2	2	2	-	-	1	-	-	-	1	2	1	1	1	1	1	1
		C205.2	Determine an appropriate process model depending on the user requirements	2	2	2	-	-	1	-	-	-	1	2	1	1	1	1	1	1
		C205.3	Identify the suitable software design with appropriate user and component level design.	2	2	2	-	-	1	-	-	-	1	2	1	1	1	1	1	1
		C205.4	Apply a suitable testing strategy for the designed software product.	2	2	2	-	-	1	-	-	-	1	2	1	1	1	1	1	1
		C205.5	Apply appropriate project estimation techniques and manage the risk.	2	2	2	-	-	1	-	-	-	1	2	1	1	1	1	1	1
6	IT1302 -Object Oriented Programming Laboratory	C205																		
		C206.1	Develop and implement Java programs for simple applications that make use of classes, packages, inheritance and interfaces.	3	3	2	-	-	1.00	3	-	-	-	1	1	1.00	1.00	1.00	2	2
		C206.2	Develop and implement Java programs with array list and exception handling	3	2	2	-	-	2	-	-	-	2	2	2	-	-	2	2	2
		C206.3	Develop and implement Java programs using the concept of Multithreading.	2	3	3	-	-	3	-	-	-	-	1	1	-	-	3	2	2
		C206.4	Design and develop the applications using file processing, generic programming.	3	3	2	-	-	2	-	-	-	2	2	2	-	-	2	2	2
7	CS1303 - Data Structures Laboratory	C206.5	Design and develop the applications using event handling mechanism.	3	3	2	-	-	3	-	-	-	1	2	2	-	-	3	2	2
		C206																		
		C207.1	Demonstrate appropriate operations and applications of Linear datastructures(List, Stack and Queue) using array	2.80	2.80	2.20	-	-	2.60	-	-	-	-	1.40	1.60	-	2.40	2.00	2.00	2.00
		C207.2	Examine functions to implement operations of Linear datastructures(List, Stack and Queue) using Linked List	3	3	2	-	-	-	-	-	-	2	1	-	-	2	3	3	2
		C207.3	Determine the different operations of various Trees	3	3	2	-	-	-	-	-	-	2	1	-	-	2	3	3	2
		C207.4	Apply graph traversal algorithms and techniques	3	3	2	-	-	-	-	-	-	2	1	-	-	2	3	3	2
		C207.5	Examine various sorting, searching and hashing algorithms	3	3	2	-	-	-	-	-	-	2	1	-	-	2	3	3	2
		C207																		
		C208.1	Illustrate the communication skills in articulating a complete idea with a clarity in pronunciation	3.00	3.00	2.00	-	-	-	-	-	-	2.00	1.00	-	-	2.00	3.00	2.00	2.00
				3	2	1	3	2	-	-	-	-	3	1	3	3	2	-	-	-

HS1301 - Interpersonal Skills Laboratory	C208.2	Demonstrate Interpersonal skills with a willingness to work in team following the social work ethics	3	1		3	2	2	3	3	3	3	3	2	-	-
	C208.3	Relate the speaking ability to respond in any informal talk detailing the content of the information	3	1		3		1	3	1	3	3	3	2	-	-
	C208.4	Participate in group discussions expressing active listening and reading skills to persuade the group to the desired target	1	3	1	3		2	3	2	3	3	3	2	-	-
	C208.5	Make effective formal presentations and interactive communication in any academic and business	2	3	2	3	3	1	3	2	3	3	3	1	-	-
	C2108		2.40	2.00	0.80	3.00	1.00	1.60	3.00	1.80	3.00	3.00	3.00	1.80	-	-

IV SEMESTER (2021-2022)EVEN

IV SEMESTER (2021-2022)EVEN																		
9	MA1403 - Probability and Statistics	C209 1	Understand the fundamental knowledge of the concepts of probability and have knowledge of standard distributions which can describe real life	3	2	3	3	-	-	-	-	-	-	-	1	2	-	
		C209 2	Understand the basic concepts of one and two dimensional random variables and apply in engineering applications.	2	2	2	2	-	-	-	-	-	-	-	-	1	2	-
		C209 3	Apply the concept of testing of hypothesis for small and large samples in real life problems.	2	3	2	1	-	-	-	-	-	-	-	-	1	2	-
		C209 4	Apply the basic concepts of classifications of design of experiments in the field of agriculture and statistical quality control.	3	3	1	1	-	-	-	-	-	-	-	-	1	2	-
		C209 5	Have the notion of sampling distributions and statistical techniques used in engineering	3	3	1	1	-	-	-	-	-	-	-	-	1	2	-
2	CS1401- Computer Architecture	C209			2.60	2.60	1.80	1.60	-	-	-	-	-	-	1.00	2.00	-	
		C210 1	Examine the basic structure of computers, operations and instructions.	3	3	2	-	-	-	-	-	-	-	-	-	2	1	2
		C210 2	Analyze the Design of the arithmetic and logic unit.	2	3	2	-	-	-	-	-	-	-	-	-	2	2	2
		C210 3	Categorize the pipelined execution and design of control unit	2	2	2	-	-	-	-	-	-	-	-	-	2	2	1
		C210 4	Illustrate the parallel processing architectures	2	3	2	-	-	-	-	-	-	-	-	-	2	1	1
		C210 5	Classify the various memory systems and I/O communication.	2	2	2	-	-	-	-	-	-	-	2	2	2		
		C210			2.20	2.60	2.00	-	-	-	-	-	-	-	2.00	1.60	1.60	
		C211 1	Model an application's data requirements using conceptual modeling and design database schemas based on the conceptual model and Formulate solutions to a broad range of query	3	2	3	-	-	-	-	-	-	-	-	2	2	2	

6	CS1403 - Object Oriented Analysis and Design	C214.2	Demonstrate the software applications using Object Oriented concepts	2	3	2	2	3	-	-	-	2	0	1	2	2	2
		C214.3	Determine the various scenarios based on software requirements	2	2	2	3	2	-	-	-	2	2	1	2	2	2
		C214.4	Experiment the UML based software design into pattern based design using design patterns	3	3	2	3	2	-	-	-	2	1	1	2	2	2
		C214.5	Illustrate the various testing methodologies for Object Oriented software	3	2	2	3	3	-	-	-	2	2	2	1	2	1
		C214															
7	IT1402 - Database Management Systems Laboratory	C215.1	Use typical data definitions and manipulative commands	3	3	3	-	1	-	-	-	2	1	-	2	2	2
		C215.2	Design applications to test Nested and Join Queries	3	3	3	-	1	-	-	-	2	1	-	2	1	3
		C215.3	Implement simple applications that use Views, Sequences, Synonyms, Cursors and critically analyze the use of Tables, Functions, Procedures, Triggers and Exception Handling using PL SQL	3	3	3	-	1	-	-	-	2	1	-	3	3	3
		C215.4	Create and manipulate data using NOSQL database	3	3	3	-	2	-	-	-	3	2	-	2	1	2
		C215.5	Design database using ER modelling, Normalize and Implement a GUI application that require a Front-end and Back end Tool	3	3	3	-	2	-	-	-	3	3	-	3	3	3
8	CS1404 - Design and Analysis of Algorithms Laboratory	C215															
		C216.1	Interpret algorithms for given problem and trace its complexity.	3	3	1	-	-	-	-	-	2	1	-	2	2	3
		C216.2	Understand the importance of different algorithm for solving same problem	3	3	3	-	-	-	-	-	2	1	-	1	2	3
		C216.3	Analyze efficient algorithms to get optimal solution for the problem	3	3	1	-	-	-	-	-	2	1	-	2	2	2
		C216.4	Apply the suitable algorithm for the given problem using backtracking method.	3	3	2	-	-	-	-	-	2	1	-	1	2	3
9	HS1401 - Employability	C216.5	Understand the approximation algorithms	3	3	2	-	-	-	-	-	2	1	-	2	2	3
		C216															
		C217.1	Write for different purposes in general and technical context	3	2	1	3	-	2	3	1	3	3	3	2	-	-
		C217.2	Write formal job applications	3	1	-	3	2	2	3	3	3	3	3	2	-	-
		C217.3	Excel in Verbal aptitude, read and evaluate texts logically to solve the puzzles.	3	1	-	3	-	1	3	1	3	3	3	2	-	-

Anal. Soft. Maths Laboratory	C201.6	Develop and demonstrate the employability and soft skills	1	3	1	3	-	2	3	2	3	3	3	3	2	-	-
	C201.5	Develops critical thinking in various professional scenarios	2	3	2	3	3	1	3	2	3	3	3	3	1	-	-
	C201.7		2.4	2.0	1.3	3.0	2.5	1.6	3.0	1.8	3.0	3.0	3.0	3.0	1.8	-	-
V SEMESTER (2022-2023)0000																	
MCA1501 - Algebra and Number Theory	C301.1	Applies the basic notions of groups, rings, fields which will then be used to solve related problems	3	3	2	-	-	-	-	-	-	-	-	-	-	2	-
	C301.2	Explains the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts	3	3	2	-	-	-	-	-	-	-	-	-	-	2	-
	C301.3	Demonstrate accurate and efficient use of advanced algebraic techniques	3	3	2	-	-	-	-	-	-	-	-	-	-	2	-
	C301.4	Demonstrate their mastery by solving non-trivial problems related to the concepts, and by proving simple theorems about the statements proven by the	3	3	2	-	-	-	-	-	-	-	-	-	-	2	-
	C301.5	Applies integrated approach to number theory and abstract algebra, and provide a firm basis for further reading and study in the subject	3	3	2	-	-	-	-	-	-	-	-	-	-	2	-
ITI1501 - Computer Networks	C301		3.00	3.00	2.00	-	-	-	-	-	-	-	-	-	-	2.00	-
	C302.1	Describe the basics of the network and discuss the functions of the physical layer	3	2	2	-	-	-	-	-	-	-	-	-	1	2	1
	C302.2	Summarize the basics of data flow from one node to another	2	2	2	-	-	-	-	-	-	-	-	-	2	1	1
	C302.3	Demonstrate the various services and protocols of the network layer	2	2	2	-	-	-	-	-	-	-	-	-	1	1	2
	C302.4	Determine the services of different transport layer protocols	2	2	1	-	-	-	-	-	-	-	-	-	2	2	1
C302	C302.5	Examine the working of various application layer protocols	2	2	1	-	-	-	-	-	-	-	-	-	2	1	1
	C302		2.20	2.00	1.60	-	-	-	-	-	-	-	-	-	1.60	1.40	1.20
	C303.1	Understand the basic concepts, functions, processes and threads of operating systems	3	2	1	-	-	-	-	-	-	-	-	-	2	2	2
	C303.2	Analyze various scheduling algorithms, Understand deadlock, prevention and avoidance algorithms	3	3	2	-	-	-	-	-	-	-	-	-	2	1	3
	C303																

3	CS1501 - Operating Systems	C303.1	Compare and contrast various memory management schemes.	3	3	2	-	-	-	-	-	-	-	-	-	2	1	2	2
		C303.4	Illustrate the functionality of file systems.	3	1	1	-	-	-	-	-	-	-	-	-	2	1	2	2
		C303.5	Perform administrative tasks on Linux Servers and compare iOS and Android Operating Systems.	3	2	1	-	-	-	-	-	-	-	-	-	2	2	2	2
		C303			3.00	2.20	1.40	-	-	-	-	-	-	-	-	2.00	1.40	2.20	2.20
		C304.1	Construction Of DFA, NFA and r-NFA and Regular expression for the languages	3	2	2	-	-	-	-	-	-	-	-	-	2	2	1	1
4	CS1502 - Theory of Computation	C304.2	Understand the concept of converting NFA to DFA, r-NFA to DFA, automata to regular expressions and regular expression to automata	2	1	1	-	-	-	-	-	-	-	-	-	2	2	1	1
		C304.3	Remembering pumping lemma for regular languages and context free languages	1	1	1	-	-	-	-	-	-	-	-	-	2	2	2	2
		C304.4	Construct a CFG, Parse trees and PDA.	1	2	2	-	-	-	-	-	-	-	-	-	2	2	1	1
		C304.5	Understand the working Principles of Turing Machine and Post correspondence problem	2	2	1	-	-	-	-	-	-	-	-	-	2	2	1	1
		C304			1.40	1.60	1.40	-	-	-	-	-	-	-	-	2.00	2.00	1.20	1.20
5	EC1515 - Embedded System	C305.1	Describe the architecture and programming of ARM processor	3	3	3	-	-	-	-	-	-	-	-	-	2	2	1	1
		C305.2	Outline the concepts of embedded systems	3	3	3	2	2	-	-	-	2	1	2	2	2	2	2	2
		C305.3	Explain the basic concepts of real time operating system design	3	3	3	-	1	-	-	-	2	1	2	2	2	2	2	2
		C305.4	Use the system design techniques to develop software for embedded systems	3	3	3	1	1	-	-	-	1	2	2	2	2	2	2	2
		C305.5	Differentiate between the general purpose operating system and the real time operating system	3	3	3	2	2	-	-	-	2	2	2	2	2	2	2	2
6	IT1503 - Networks	C305			3.00	3.00	3.00	1.67	1.50	-	-	-	1.75	1.50	2.00	2.00	1.50	1.50	1.50
		C306.1	Illustrate the use of TCP and UDP protocols.	3	2	2	-	3	-	-	-	2	-	-	-	2	2	2	2
		C306.2	Examine the performance of different transport layer protocols.	3	2	2	-	2	-	-	-	1	-	-	-	2	2	1	1
		C306.3	Use simulation tools to determine the performance of various network protocols.	2	2	2	-	3	-	-	-	2	-	-	-	2	2	2	2

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3	CS1603 - Artificial Intelligence	C309															
		C310.1	Illustrate the intelligent agent frameworks and its environments.	2	2	1	1	-	-	-	-	-	-	-	-	3.00	2.20
		C310.2	Analyze real life problems and solve those using searching algorithms and AI techniques.	3	3	2	2	-	-	-	-	-	-	2	2	2	1
		C310.3	Analyze and apply logical reasoning in solving AI problems using first order and predicate logic.	3	3	2	2	-	-	-	-	-	-	2	3	2	2
		C310.4	Demonstrate the applications of Probabilistic Reasoning in Bayesian Networks.	3	3	2	2	-	-	-	-	-	-	1	2	2	2
		C310.5	Examine the various applications of AI and types of Learning.	2	1	2	2	-	-	-	-	-	-	1	2	2	1
4	CS1604 - Cryptography and Network Security	C310															
		C311.1	Understand the fundamentals of network security, security architecture, threats, and vulnerabilities	2	2	2	2	-	-	-	-	-	-	1	2	-	3
		C311.2	Apply symmetric cryptographic algorithms to perform encryption and decryption	3	2	2	2	-	-	-	-	-	-	1	2	-	3
		C311.3	Demonstrate public-key cryptographic operations for secure communication	3	2	3	3	-	-	-	-	-	-	2	1	-	3
		C311.4	Implement and analyze authentication mechanisms to ensure secure access control in information systems.	3	1	2	2	-	-	-	-	-	-	2	2	-	3
		C311.5	Experiment with advanced encryption methods, including identity-based encryption and quantum cryptography techniques	2	2	2	3	-	-	-	-	-	-	2	2	-	3
5	CS1607 - Cyber Security	C311															
		C312.1	Acquire the basic concepts of Cyber Security.	3	3	3	3	3	2	2	1	2	2	2	2	2	3
		C312.2	Determine the security applications for operating system and networks	2	3	2	1	1	2	2	1	2	2	2	2	3	3
		C312.3	Discover the security detection and classify their various security countermeasures.	2	3	2	2	3	2	2	2	1	1	2	1	2	3
		C312.4	Infer the various applicable Privacy principles and policies	2	2	2	1	1	3	2	2	1	1	1	1	2	3
		C312.5	Identify security planning and effective management of cyber security	2	2	2	2	1	2	2	1	1	1	1	1	3	3

			C312																
			C312.1	Understand the fundamentals of software testing															
			C312.2	Design test cases suitable for a software development for different domains															
			C312.3	Design test cases and document test plan															
			C312.4	Apply a wide variety of testing techniques in an effective and efficient manner															
			C312.5	Apply automatic testing tools in different domains															
			C313																
			C313.1	Design and Implement Web pages using HTML/XML and style sheets.															
			C313.2	Develop Client side dynamic web pages with validation using Java Script and JSON															
			C313.3	Develop Server side dynamic web pages with validation using Servlet and JSP															
			C313.4	Construct simple dynamic and interactive Web pages using PHP.															
			C313.5	Develop interactive web applications using recent Tools															
			C314																
			C314.1	Develop code for classical Encryption Techniques to solve the problems															
			C314.2	Build cryptosystems by applying symmetric and public-key encryption algorithms.															
			C314.3	Construct code for authentication algorithms.															
			C314.4	Develop a signature scheme using the Digital Signature Standard.															
			C314.5	Demonstrate the network security system using open-source tools.															
			C315																
			C315.1	Listen and respond appropriately															
			C315.2	Participate in group discussions															

9	HS1601 – Professional Communication Laboratory	C316.3	Make effective presentations	-	-	-	-	-	3	3	3	3	3	3	3	3	3	3	-	-	-
		C316.4	Participate confidently and appropriately in conversations both formal and informal	-	-	-	-	-	3	3	3	3	3	3	3	3	3	3	-	-	-
		C316.5	Persuade through conversations.	-	-	-	-	-	3	3	3	3	3	3	3	3	3	3	-	-	-
		C316			-	-	-	-	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	-	-	-
VII SEMESTER (2023-2024)EVEN																					
1	IT1701- Distributed Systems and Cloud Computing	C401.1	Describe the concepts of distributed systems and Remote procedure calls.	3	2	3	-	-	3	-	1	2	2	2	2	2	2	2	2	2	2
		C401.2	Able to understand real time distributed systems and load balancing approaches.	3	2	2	-	-	3	-	2	-	2	2	2	2	2	2	2	2	2
		C401.3	Elucidate the concepts of NIST Cloud Computing architecture and its design challenges.	2	2	3	-	-	2	-	2	2	2	2	2	2	2	2	2	2	2
		C401.4	Implement different types of Virtualization technologies.	3	2	2	-	-	3	-	2	2	2	1	2	2	2	2	2	2	2
		C401.5	Analyse the issues in Resource provisioning and Security governance in clouds.	2	2	3	-	-	3	-	1	-	1	2	2	2	2	2	2	2	2
2	CS1701-Machine Learning	C401			2.60	2.00	2.60	-	2.80	-	1.60	2.00	1.60	2.00	2.00	2.00	2.00	2.00	2.00	2.00	2.00
		C402.1	Understand the basic concepts of machine learning and its types	2	1	1	1	1	-	-	-	-	-	-	-	-	-	-	1	1	1
		C402.2	Solve the problems of different applications using supervised learning algorithms	2	2	2	2	2	1	-	-	-	-	-	-	-	-	-	1	2	2
		C402.3	Analyze typical Unsupervised learning algorithms for different types of applications.	2	2	2	2	2	1	-	-	-	-	-	-	-	-	-	1	2	2
		C402.4	Demonstrate the various Reinforcement algorithms	2	2	2	2	2	1	-	-	-	-	-	-	-	-	-	1	2	2
3	MG1701- Principles of	C402.5	Examine probabilistic methods for real world applications.	2	2	2	2	2	-	-	-	-	-	-	-	-	-	1	2	2	
		C402			2.00	1.80	1.80	1.80	1.80	1.00	-	-	-	-	-	-	-	1.00	1.80	1.80	1.80
		C403.1	Understand about the planning and decision making process of management.	1	2	1	-	-	-	2	1	-	-	-	-	-	-	-	2	2	2
		C403.2	Understand the organization structure, HR planning and control.	2	2	2	-	-	-	2	1	-	1	1	1	1	2	2	2	2	2
		C403.3	Apply functions of management like motivation, leadership and communication.	2	1	1	1	-	-	-	2	-	-	1	-	-	2	2	2	2	2

4	Management	C403.4	Analyze the performance controlling process, techniques of control and reporting to the	2	2	1	-	-	2	1	1	1	1	-	2	2	2	1
		C403.5	Apply principles of management in order to exercise the role as a manager in IT industry.	2	2	2	-	-	2	1	1	1	1	2	2	2	2	2
		C403																
		C404.1	Learn the effective dialog for HCI	2	3	2	2	3	2	2	1	1	1	1	2	2	2	3
		C404.2	Design effective HCI for individuals and persons with disabilities	2	2	2	2	3	2	2	1	2	1	2	1	2	2	2
5	CS1707-Human Computer Interactions	C404.3	Understand the importance of user feedback.	2	2	2	2	3	2	1	1	1	1	1	2	2	2	1
		C404.4	Comprehend the HCI implications for designing multimedia, e-commerce/ e-learning Websites.	2	1	2	3	2	3	3	1	1	2	1	2	1	2	2
		C404.5	Develop meaningful user interface.	2	1	2	3	2	2	1	1	1	1	2	1	1	1	2
		C404																
		C405.1	Develop a simple applications using C# and describe the .Net framework	2	2	2	3	3	-	-	-	-	-	-	-	-	3	2
6	CS1708-C# and .Net Programming	C405.2	Develop various applications using C# advanced features in .Net Framework.	2	3	2	3	3	-	-	-	-	-	-	2	3	3	2
		C405.3	Develop distributed applications by implementing base class libraries and database connectivity using .Net Framework.	2	2	2	3	2	2	-	-	-	-	-	2	2	2	2
		C405.4	Develop window based applications using .Net Framework.	2	2	2	2	3	-	-	-	-	-	-	2	2	2	2
		C405.5	Design a mobile application using .NET compact framework.	2	2	2	2	2	-	-	-	-	-	-	2	2	2	2
		C405																
7	C#1711 -Full Stack Web Development	C406.1	Utilize CSS and JavaScript for effective client-side scripting.	2	2	2	-	3	2	2	-	2	2	2	3	3	3	2
		C406.2	Design and implement the client-side architecture of web applications.	2	2	2	-	2	2	2	-	2	2	2	3	3	3	1
		C406.3	Develop web applications using NoSQL.	2	2	2	-	3	2	2	-	2	2	2	3	3	3	2
		C406.4	Design and manage NoSQL databases with MongoDB.	2	2	2	-	2	2	2	-	3	2	2	3	3	3	2
		C406.5	Build and deploy a full-stack Single Page Application (SPA) utilizing React, NodeJS, and MongoDB in a cloud environment.	2	2	2	-	3	2	2	-	1	2	2	3	3	3	2

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2	CS1809-UI & UX Design	C413.4	Outline Wireframes, Prototypes, and Usability Tests																3	1	2	3	-	-	-	-	-	-	-	-	-	1	2
		C413.5	Apply Research, Ideation, and Information Architecture																3	1	1	3	-	-	-	-	-	-	-	-	-	1	2
		C413																2.40	1.60	1.80	2.40	-	-	-	-	-	-	-	-	1.60	1.60		
3	CS1810-Green Computing	C414.1	Acquire knowledge to adopt green computing practices to minimize negative impacts on the environment.																2	2	2	2	1	2	3	-	2	1	1	2	2	2	
		C414.2	Understand the green assets and modeling.																2	2	1	1	2	2	2	-	-	1	1	1	2	2	
		C414.3	Enhance the skill in energy saving practices in their use of hardware.																3	2	2	2	1	2	1	-	1	1	1	2	2		
		C414.4	Evaluate technology tools that can reduce paper waste and carbon footprint by the stakeholders.																2	2	1	1	-	2	2	-	-	-	-	2	2		
		C414.5	Understand the ways to minimize equipment disposal requirements.																3	2	2	2	-	2	2	-	-	-	-	1	2	2	
		C418																2.4	2	1.6	1.6	1.3333	2	2	-	1.5	1	1	1.4	2	2		
4	CS1813-Digital Marketing	C415.1	Understand about the designing, developing and promoting through Blogs																2	1	2	-	3	-	-	-	3	-	-	1	1		
		C415.2	Apply the search engine techniques for better visibility in the online world																2	1	1	-	3	-	-	-	3	-	-	2	2	2	
		C415.3	Develop an e-mail / Mobile marketing for promotional activities																2	2	2	-	3	-	-	-	3	-	-	2	1	1	
		C415.4	Apply social media marketing techniques to build customer relationships.																2	2	2	-	3	-	-	-	3	-	-	2	2	1	
		C415.5	Apply digital transformation techniques in email,mobile and social media																2	2	2	-	3	-	-	-	3	-	-	1	1	2	
		C415																2	1.6	1.8	-	3	-	-	-	3	-	-	1.6	1.4	1.4		

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S.A ENGINEERING COLLEGE, CHENNAI - 77
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
CO-PO-PSO Mapping

Batch:2019-2023

S.NO	SUB.CODE& NAME	COURSE OUTCOME S	PROGRAMME OUTCOMES												PSO		
			PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
			I SEMESTER (2019-2020) ODD														
1	HS8151 Communicative English	C101.1	-	-	-	-	-	-	-	2	3	3	-	2	-	-	-
		C101.2	-	-	-	-	-	-	-	2	3	3	-	2	3	3	-
		C101.3	-	-	-	-	-	-	-	2	3	3	-	2	-	-	-
		C101.4	-	-	-	-	-	-	-	2	3	3	-	2	-	-	-
		C101.5	-	-	-	-	-	-	-	2	3	3	-	2	-	1	-
		C101.6	-	-	-	-	-	-	-	2	3	3	-	2	1	1	-
		C101	-	-	-	-	-	-	-	2.00	3.00	3.00	-	2.00	2.00	1.67	-
2	MA8151 Engineering Mathematics I	C102.1	3	3	3	-	-	-	-	-	2	-	-	-	3	-	-
		C102.2	3	3	3	-	-	-	-	-	2	-	-	-	1	-	-
		C102.3	3	3	3	-	-	-	-	-	2	-	-	-	-	-	-
		C102.4	3	3	3	-	-	-	-	-	2	-	-	-	1	-	-
		C102.5	3	3	3	-	-	-	-	-	2	-	-	-	2	2	-
		C102.6	3	3	3	-	-	-	-	-	2	-	-	-	2	2	-
		C102	3.00	3.00	3.00	-	-	-	-	-	2.00	-	-	-	1.80	2.00	-
3	PH8151 Engineering Physics	C103.1	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-
		C103.2	3	3	3	-	-	-	-	-	-	-	-	-	3	3	-
		C103.3	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-
		C103.4	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-
		C103.5	3	3	3	-	-	-	-	-	-	-	-	-	-	1	-
		C103.6	3	3	3	-	-	-	-	-	-	-	-	-	1	1	-
		C103	3.00	3.00	3.00	-	-	-	-	-	-	-	-	-	2.00	1.67	-
4	CY8151 Engineering Chemistry	C104.1	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.2	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.3	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.4	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.5	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.6	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104	3.00	2.00	2.00	-	-	-	-	-	-	-	-	-	-	-	-
5	GE8151 Problem Solving and Python Programming	C105.1	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.2	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.3	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.4	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.5	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.6	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105	3.00	2.00	2.00	-	-	-	-	-	-	-	-	-	1.00	-	1.00
		C106.1	3	3	2	-	2	-	-	3	3	2	-	2	1	-	

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8	MOBILE APPLICATION DEVELOPMENT LABORATORY	C317.4	2	1	3	-	-	3	2	-	-	-	-	1	3	1	-	2	2	-	2	2	-	2	2	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
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3	COMPUTING	C403	2.17	2.67	3.00	-	-	-	-	-	-	-	2.50	2.67	2.33
4	OMIE752/SUPPLY CHAIN MANAGEMENT	C404.1	1	-	-	-	1	1	1	1	1	1	1	1	1
		C404.2	1	-	-	-	1	1	1	1	1	1	1	1	1
		C404.3	2	-	-	-	1	1	1	1	1	1	1	1	1
		C404.4	1	-	-	-	1	1	1	1	1	1	1	1	1
		C404.5	1	-	-	-	1	1	1	1	1	1	1	1	1
		C404.6	1	-	-	-	1	1	1	1	1	1	1	1	1
5	T	C404	1.17	-	-	-	1.00	1.00	1.00	1.00	1.00	2.00	1.83	2.17	1.50
		C405.1	2	-	-	-	1	1	1	1	1	1	-	-	-
		C405.2	1	-	-	-	1	1	1	1	1	1	-	-	-
		C405.3	2	-	-	-	1	1	1	1	1	1	-	-	-
		C405.4	1	-	-	-	1	1	1	1	1	1	1	-	-
		C405.5	2	-	-	-	1	1	1	1	1	1	-	-	-
5	OBM752/HOSPITAL MANAGEMENT	C405.6	-	-	-	-	2	2	2	2	2	2	-	-	-
		C405	1.60	-	-	-	1.17	1.67	1.00	1.00	1.00	1.00	1.00	-	-
		C406.1	3	3	3	3	1	1	1	1	1	1	3	3	2
		C406.2	3	3	2	1	2	-	-	-	-	-	3	3	2
		C406.3	3	3	3	2	1	-	-	-	-	-	3	3	-
		C406.4	1	1	-	-	-	-	-	-	-	-	3	3	-
5	Professional Readiness for Innovation, Employability and Entrepreneurship	C406.5	3	3	3	3	2	1	1	1	1	2	3	2	2
		C406.6	3	2	2	1	1	1	1	1	1	1	2	-	-
		C406	2.67	2.50	2.00	1.50	1.00	1.00	1.00	1.00	1.00	2.50	2.60	2.25	2.00
		C407.1	2	-	-	3	-	-	-	-	-	2	2	-	-
		C407.2	2	-	-	-	-	-	-	-	-	-	2	-	-
		C407.3	2	-	-	3	-	-	-	-	-	2	2	-	-
6	CS873/C# AND .NET PROGRAMMING	C407.4	2	-	-	3	-	-	-	-	-	1	2	-	-
		C407.5	2	-	-	-	-	-	-	-	-	1	2	-	-
		C407.6	2	-	-	3	-	-	-	-	-	2	2	-	-
		C407	2.00	-	-	3.00	-	-	-	-	-	1.60	2.00	-	1.00
		C408.1	3	-	-	-	-	-	-	-	-	-	1	-	-
		C408.2	3	2	-	-	-	-	-	-	-	-	-	2	-
7	CS8079/HUMAN COMPUTER INTERACTION	C408.3	3	-	-	-	-	-	-	-	-	-	2	-	-
		C408.4	3	2	-	-	-	-	-	-	-	-	-	2	-
		C408.5	3	2	-	-	-	-	-	-	-	-	-	2	-
		C408.6	3	2	-	-	-	-	-	-	-	-	-	2	-
		C408	3	2	-	-	-	-	-	-	-	-	-	2	-
		C409.1	3	2	-	-	-	-	-	-	-	-	-	2	-
8	CLOUD COMPUTING LABORATORY	C409.2	2	2	-	2	-	-	-	-	-	2	1.5	2	-
		C409.3	3	2	-	3	-	-	-	-	-	1	2	3	2
		C409.4	2	2	-	2	-	-	-	-	-	1	3	3	2
		C409.5	3	3	-	3	-	-	-	-	-	1	2	2	2
		C409.6	2	2	-	2	-	-	-	-	-	2	2	2	1
		C409	2.50	2.17	2.33	-	-	-	-	-	-	1.33	2.17	2.67	1.83
		C410.1	3	3	-	2	-	-	-	-	-	1	2	2	3
		C410.2	3	3	-	2	-	-	-	-	-	1	1	2	1
		C410.3	3	3	-	2	-	-	-	-	-	2	3	1	2



2018-2022

**MAPPING OF COURSE
OUTCOMES WITH
PROGRAM
OUTCOMES**

S.A ENGINEERING COLLEGE, CHENNAI - 77
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
CO-PO-PSO Mapping
Batch:2018-2022

S.NO	SUB.CODE& NAME	COURSE OUTCOMES	PROGRAMME OUTCOMES												PSO		
			PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
			I SEMESTER (2018-2019) ODD														
1	HS8151 Communicative English	C101.1	-	-	-	-	-	-	-	2	3	3	-	2	-	-	-
		C101.2	-	-	-	-	-	-	-	2	3	3	-	2	3	3	-
		C101.3	-	-	-	-	-	-	-	2	3	3	-	2	-	-	-
		C101.4	-	-	-	-	-	-	-	2	3	3	-	2	-	-	-
		C101.5	-	-	-	-	-	-	-	2	3	3	-	2	-	1	-
		C101.6	-	-	-	-	-	-	-	2	3	3	-	2	1	1	-
		C101	-	-	-	-	-	-	-	2.00	3.00	3.00	-	2.00	2.00	1.67	-
2	MA8151 Engineering Mathematics I	C102.1	3	3	3	-	-	-	-	-	2	-	-	-	3	-	-
		C102.2	3	3	3	-	-	-	-	-	2	-	-	-	1	-	-
		C102.3	3	3	3	-	-	-	-	-	2	-	-	-	-	-	-
		C102.4	3	3	3	-	-	-	-	-	2	-	-	-	1	-	-
		C102.5	3	3	3	-	-	-	-	-	2	-	-	-	2	2	-
		C102.6	3	3	3	-	-	-	-	-	2	-	-	-	2	2	-
		C102	3.00	3.00	3.00	-	-	-	-	-	2.00	-	-	-	1.80	2.00	-
3	PH8151 Engineering Physics	C103.1	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-
		C103.2	3	3	3	-	-	-	-	-	2	-	-	-	3	3	-
		C103.3	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-
		C103.4	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-
		C103.5	3	3	3	-	-	-	-	-	-	-	-	-	-	1	-
		C103.6	3	3	3	-	-	-	-	-	-	-	-	-	1	1	-
		C103	3.00	3.00	3.00	-	-	-	-	-	-	-	-	-	2.00	1.57	-
4	CY8151 Engineering Chemistry	C104.1	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.2	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.3	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.4	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.5	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104.6	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-
		C104	3.00	2.00	2.00	-	-	-	-	-	-	-	-	-	-	-	-
5	GE8151 Problem Solving and Python Programming	C105.1	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.2	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.3	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.4	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.5	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105.6	3	2	2	-	-	-	-	-	-	-	-	-	1	-	1
		C105	3.00	2.00	2.00	-	-	-	-	-	-	-	-	-	1.00	-	1.00
6	GE8152 Engineering Graphics	C106.1	3	3	2	-	2	-	-	3	3	2	-	2	2	1	-
		C106.2	3	3	2	-	2	-	-	3	3	2	-	2	2	1	-
		C106.3	3	3	2	-	2	-	-	3	3	2	-	2	2	1	-
		C106.4	3	3	2	-	2	-	-	3	3	2	-	2	2	1	-
		C106.5	3	3	2	-	2	-	-	3	3	2	-	2	2	1	-
		C106.6	3	3	2	-	2	-	-	3	3	2	-	2	2	1	-
		C106	3.00	3.00	2.00	-	2.00	-	-	3.00	3.00	2.00	-	2.00	2.00	1.17	-

8	Laboratory	C308	2.17	2.67	2.83	-	2.67	2.67	-	2.00	2.67	2.50	-	2.00	2.33	-	-
9	CS8581 Networks Laboratory	C309.1	2	1	1	-	-	-	-	1	1	1	-	1	2	-	2
		C309.2	3	2	2	-	-	-	-	3	3	3	-	3	2	-	2
		C309.3	3	2	2	-	-	-	-	2	2	2	-	2	2	-	2
		C309.4	3	2	2	-	-	-	-	1	1	1	-	1	2	-	2
		C309.5	3	2	2	-	-	-	-	3	3	3	-	3	2	-	2
		C309.6	3	2	2	-	-	-	-	3	3	3	-	3	2	-	2
VI SEMESTER (2020-2021)EVEN																	
1	CS8651 Internet Programming	C310.1	3	2	3	-	-	-	-	1	1	1	-	2	-	-	-
		C310.2	3	2	3	-	-	-	-	1	1	1	-	2	-	-	-
		C310.3	3	2	3	-	-	-	-	1	1	1	-	2	-	-	-
		C310.4	3	2	3	-	-	-	-	1	1	1	-	2	-	-	-
		C310.5	3	2	3	-	-	-	-	1	1	1	-	2	-	-	-
		C310.6	3	2	3	-	-	-	-	1	2	2	-	2	-	-	-
2	CS8691 Artificial Intelligence	C310	3.00	2.00	3.00	-	-	-	-	1.00	1.17	1.17	-	2.00	2.00	2.00	-
		C311.1	3	3	3	-	-	-	-	-	-	-	-	-	2	-	-
		C311.2	3	3	3	-	-	-	-	-	-	-	-	-	1	-	-
		C311.3	3	3	3	-	-	-	-	-	-	-	-	-	1	-	-
		C311.4	3	3	3	-	-	-	-	-	-	-	-	-	1	-	-
		C311.5	3	3	3	-	-	-	-	-	-	-	-	-	1	-	-
3	CS8661 Mobile Computing	C311.6	3	3	3	-	-	-	-	-	-	-	-	2	2	2	-
		C311	3.00	3.00	3.00	-	-	-	-	-	-	-	-	2.00	1.33	1.33	2.00
		C312.1	3	2	1	-	-	-	-	-	-	-	-	-	-	-	-
		C312.2	3	2	1	-	-	-	-	-	-	-	-	-	-	-	-
		C312.3	3	3	1	-	-	-	-	-	-	-	-	-	-	-	-
		C312.4	3	2	1	-	-	-	-	-	-	-	-	-	-	-	-
4	CS862 Compiler Design	C312.5	3	2	3	-	-	-	-	-	-	-	-	-	-	-	-
		C312.6	3	2	3	-	-	-	-	-	-	-	-	-	-	2	1
		C312	3.00	2.17	1.67	-	-	-	-	-	-	-	-	-	-	-	1
		C313.1	3	2	2	-	-	-	-	1	1	1	-	2	-	-	-
		C313.2	2	2	2	-	-	-	-	1	1	1	-	1	-	-	-
		C313.3	2	2	3	-	-	-	-	1	2	2	-	2	-	2	-
5	CS863 Distributed Systems	C313.4	2	3	2	-	-	-	-	1	1	1	-	2	-	-	-
		C313.5	3	2	2	-	-	-	-	2	1	1	-	2	-	-	-
		C313.6	3	2	3	-	-	-	-	1	2	1	-	1	-	-	-
		C313	2.50	2.17	2.33	-	-	-	-	1.17	1.33	1.17	-	1.67	-	2.00	-
		C314.1	3	2	1	-	-	-	-	-	-	-	-	-	2	-	-
		C314.2	3	2	1	-	-	-	-	-	-	-	-	-	-	-	1
6	CS875 DATA WAREHOUSING AND DATA MINING	C314.3	3	2	1	-	-	-	-	-	-	-	-	-	-	-	1
		C314.4	3	1	1	-	-	-	-	-	-	-	-	-	-	-	1
		C314.5	3	2	1	-	-	-	-	-	-	-	-	-	-	-	1
		C314.6	1	1	1	-	-	-	-	-	-	-	-	-	-	-	-
		C314	2.67	1.67	1.00	-	-	-	-	-	-	-	-	-	-	-	-
		C315.1	3	2	-	-	-	-	-	-	-	-	-	-	2.00	-	1.00

3	CS8791/ COLUD COMPUTING	C403.2	2	3	3	-	-	-	-	-	-	-	-	-	-	-	-	3	2	2
		C403.3	2	3	3	-	-	-	-	-	-	-	-	-	-	-	-	2	3	3
		C403.4	2	3	3	-	-	-	-	-	-	-	-	-	-	-	-	2	3	2
		C403.5	2	3	3	-	-	-	-	-	-	-	-	-	-	-	-	3	2	2
		C403.6	3	1	3	-	-	-	-	-	-	-	-	-	-	-	-	3	3	2
		C403	2.17	2.67	3.00	-	-	-	-	-	-	-	-	-	-	-	-	2.50	2.67	2.33
		C404.1	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	1	1
4	OME752/SUPPLY CHAIN MANAGEMENT	C404.2	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	2	1
		C404.3	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3	3	2
		C404.4	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	2	1
		C404.5	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	3	3	3
		C404.6	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	2	1
		C404	1.17	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1.83	2.17	1.50
		C405.1	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
5	CS891/BIG DATA ANALYTICS	C405.2	2	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
		C405.3	2	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
		C405.4	2	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
		C405.5	2	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
		C405.6	2	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	-	1
		C405	2.17	2.17	2.17	-	-	-	-	-	-	-	-	-	-	-	-	2.00	-	1.00
		C406.1	3	3	3	-	-	-	-	-	-	-	-	-	-	-	-	3	2	3
6	IT875/SOFTWARE PROJECT MANAGEMENT	C406.2	3	2	3	-	-	-	-	-	-	-	-	-	-	-	-	2	3	2
		C406.3	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	2	2
		C406.4	3	2	3	-	-	-	-	-	-	-	-	-	-	-	-	3	3	3
		C406.5	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	2	3
		C406.6	3	2	2	-	-	-	-	-	-	-	-	-	-	-	-	2	2	2
		C406	3.00	2.17	2.33	-	-	-	-	-	-	-	-	-	-	-	-	2.33	2.33	2.50
		C407.1	2	-	3	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
7	CS873/C# AND .NET PROGRAMMING	C407.2	2	-	2	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
		C407.3	2	-	3	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
		C407.4	2	-	3	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
		C407.5	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	-	-
		C407.6	2	-	3	-	-	-	-	-	-	-	-	-	-	-	-	2	-	1
		C407	2.00	-	2.80	-	-	-	-	-	-	-	-	-	-	-	-	2.00	-	1.00
		C407	2.00	-	2.80	-	-	-	-	-	-	-	-	-	-	-	-	2.00	-	1.00